Ujjwal Karki

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EDUCATION

+2 Dhulabari Secondary School, Dhulabari *3.3 GPA*
BSc.IT (Presidential Graduate School, Kathmandu) 4.0 GPA

SKILLS

Python (NumPy, Pandas, Matplotlib, Seaborn, TensorFlow, Scikit-learn), Pipeline, SQL, Data Cleaning, Exploratory Data Analysis (EDA), Statistical Analysis, Supervised ML (Regression, Classification), Model Evaluation & Optimization, Git/GitHub, Jupyter Notebook, VS Code, MS Excel, FastAPI (basics), AWS EC2, Vercel

PROJECTS

Chatbot (bainiAI)

- Built an LLM-based chatbot with LangChain, Gemini, and ChromaDB capable of answering document-based queries using RAG
- Implemented a FastAPI backend and React Frontend in Vite, integrating seamless conversation flow and Excel-based appointment tracking

Crop Disease Prediction

- Developed a deep learning model to classify plant leaf diseases using CNNs; deployed backend using FastAPI + Docker; pre-processed images using augmentation to improve generalization
- Created working web app that supports real-time image upload and disease prediction using Vercel and AWS EC2

Wine Quality Prediction

- Built a classification model to predict wine quality using chemical properties, applying EDA, and feature importance analysis
- Balanced the dataset and improved recall using SMOTE and hyperparameter tuning

Boston House Prediction

- Created a regression model to predict housing price using Scikit-learn, applying feature scaling, correlation analysis, model comparison, as well as visualization using Matplotlib/Seaborn
- Improved model performance by testing Linear Regression and Random Forest while analyzing key drivers of price

Titanic Survival Prediction

- Built an end-to-end ML model predicting passenger survival using feature engineering, logistic regression, and cross-validation
- Performed data cleaning and exploratory analysis to uncover patterns in demographics, cabin class, and survival rates

ACHIVEMENTS

- Completed a multi-project ML journey, building and deploying real-world projects including a crop disease classifier and a RAG-based chatbot
- Developed end-to-end data and ML pipelines using Scikit-learn, FastAPI, Docker, etc
- Demonstrated strong consistency and self-discipline through long-term learning streaks and continuous project execution

INTERESTS

ML, AI, Coding, Beatboxing, Gaming, Content Creation, Video Editing